



Evaluating IoT protocols with the 6TiSCH simulator

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Abstract

This report presents an evaluation of the 6TiSCH network using the 6TiSCH simulator. The Time-Slotted Channel Hopping (TSCH) mode of IEEE 802.15.4e is investigated under various network conditions and configurations to analyze its performance in terms of packet delivery rate, latency, join time, synchronization time and network lifetime.

1 Introduction

The Internet of Things (IoT) is rapidly evolving, and the need for efficient, reliable wireless communication standards is paramount. The 6TiSCH working group aims to address this need by combining the industrial performance of IEEE 802.15.4 TSCH with the interoperability and flexibility of IPv6. This report explores the effectiveness of the 6TiSCH protocol stack through simulation. Prior to conducting the simulations, a virtual Python environment was created using `venv` to manage dependencies effectively. The simulator's software repository was cloned to the local machine, and the necessary requirements were installed, setting the stage for running the simulations on a personal laptop. This preparation ensured an isolated and consistent environment for a fair assessment of the Minimal Scheduling Function (MSF) and the Simple Function (SFNone).

2 Methodology

2.1 Performance Metrics

In evaluating the performance of the Minimal Scheduling Function (MSF) and the Simple Function (SFNone) using the 6TiSCH simulator, we focus on several Key Performance Indicators (KPIs). The `sf_class` parameter in the `config.json` file is set to either "MSF" or "SFNone" to select the scheduling function for simulation. These KPIs include join time, latencies, network lifetime, synchronization time, and packet delivery reliability, which together offer a comprehensive picture of the network performance.

- **Latencies:** The end-to-end delay experienced by packets, which affects the timeliness of the data transmission in IoT applications. We also compare latencies maximum and average.
- **Join Time:** The duration for a mote to join the network, which is critical for network scalability and responsiveness.
- **Network Lifetime:** Estimated lifetime of the network based on battery consumption, important for maintenance and cost.
- **Synchronization Time:** Time required for motes to synchronize with the network, influencing the efficiency of network operations.
- **Packet Delivery Reliability:** Measured by the number of successfully delivered packets versus lost packets, indicating the reliability of the network communication.

3 Results and Discussion

3.1 Comparison of KPIs:

3.1.1 Latencies

The simulation results indicate that the MSF typically outperforms the SFNone in terms of latency. This suggests a network under MSF would be more responsive, an essential aspect for real-time IoT applications.

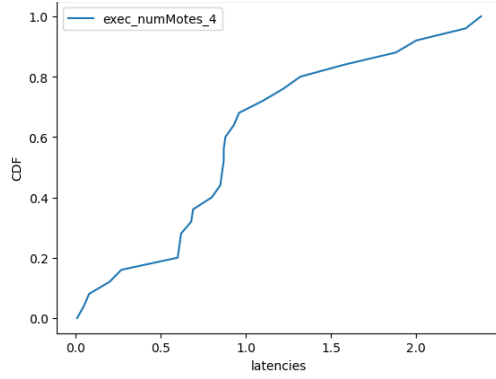


Figure 1: Latency using MSF

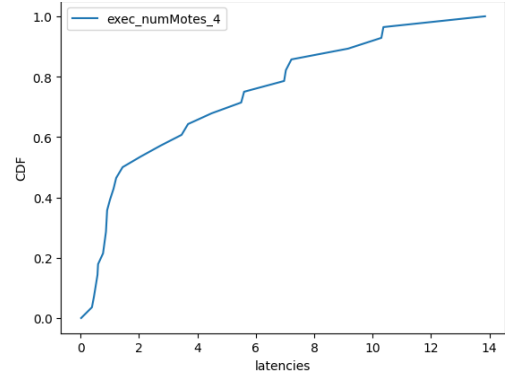


Figure 2: Latency using SFNone

3.1.2 Latencies maximum

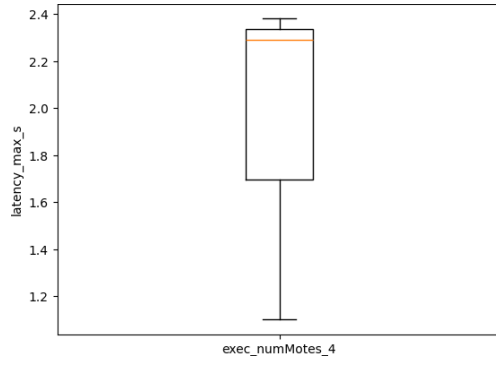


Figure 3: Latency maximum using MSF

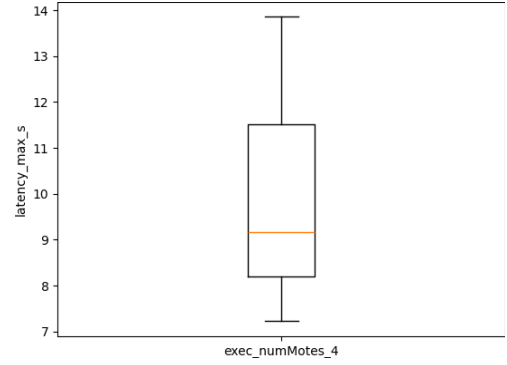


Figure 4: Latency maximum using SFNone

3.1.3 Latencies average

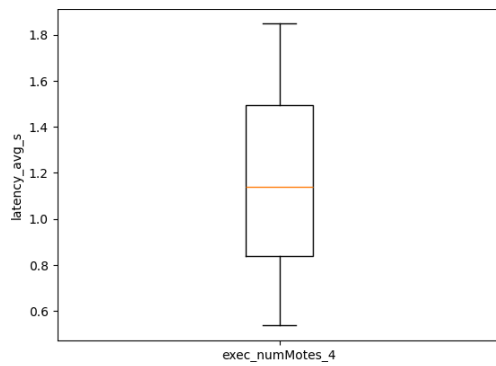


Figure 5: Latency average using MSF

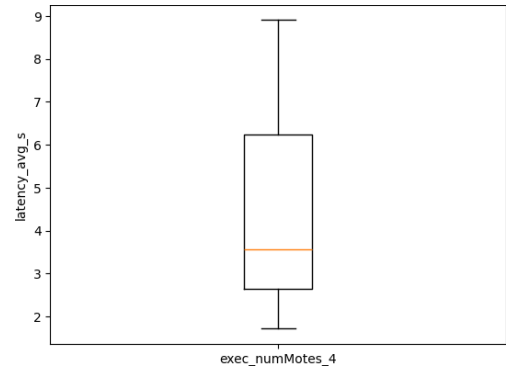


Figure 6: Latency average using SFNone

3.1.4 Join Time

Join time results highlight MSF's efficiency, allowing devices to connect faster compared to SFNone, demonstrating better scalability potential under MSF.

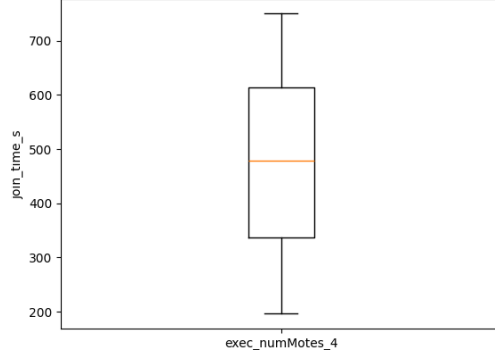


Figure 7: Join Time using MSF

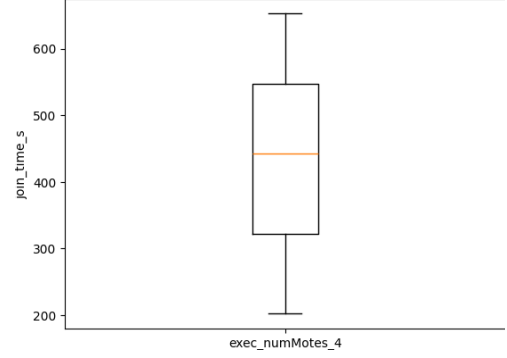


Figure 8: Join Time using SFNone

3.1.5 Network lifetime

The simulations show similar network lifetimes between MSF and SFNone, indicating that MSF's performance benefits do not come at the cost of excessive energy consumption.

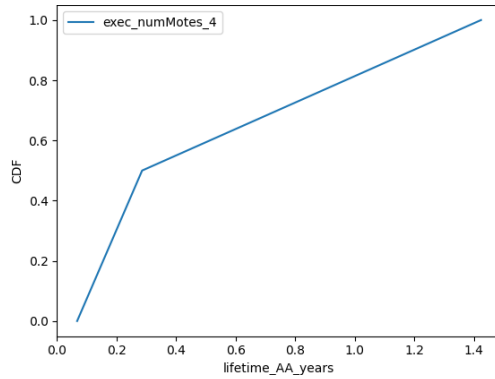


Figure 9: Network lifetime using MSF

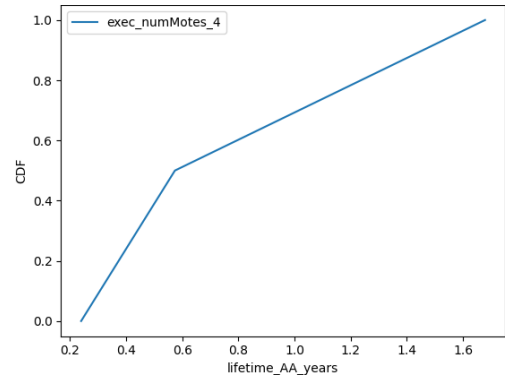


Figure 10: Network lifetime using SFNone

3.1.6 Synchronization time

Synchronization time is significantly improved with MSF, pointing to a more synchronized network operation, which can be particularly beneficial in IoT environments requiring coordinated device interactions.

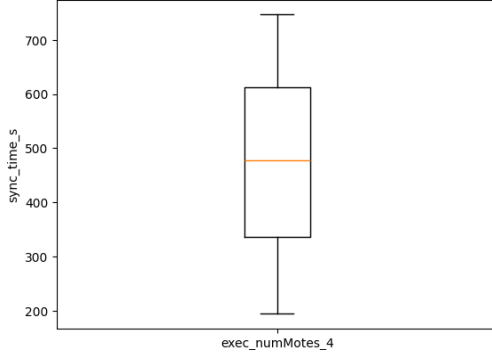


Figure 11: Synchronization time using MSF

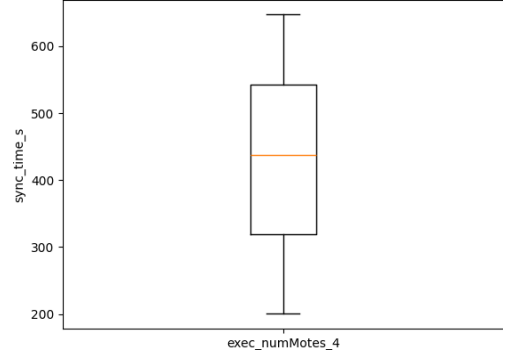


Figure 12: Synchronization time using SFNone

3.1.7 Packet delivery reliability

Packet delivery reliability data reveals fewer lost packets with MSF, emphasizing its reliability advantage for stable IoT communication networks.



Figure 13: Packet delivery reliability using MSF

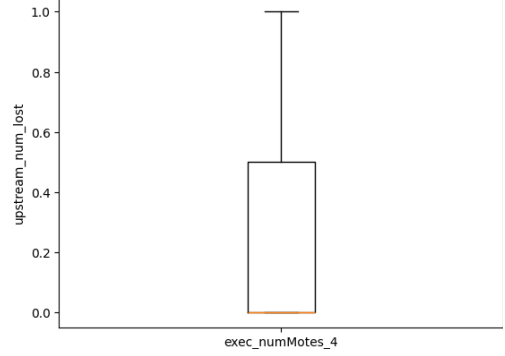


Figure 14: Packet delivery reliability using SFNone

3.2 Brief discussion

The simulation results reveal that MSF outperforms SFNone in key areas. MSF shows enhanced network responsiveness and reliability, with lower latencies and synchronization times, crucial for time-sensitive IoT applications. Importantly, these benefits are achieved without compromising network lifetime, indicating efficient energy use. MSF's robustness in packet delivery further underscores its suitability for scalable and reliable IoT networks. This performance superiority positions MSF as a preferred scheduling function for demanding IoT environments.

4 Conclusion

The simulations conducted reveal that the Minimal Scheduling Function (MSF) outperforms the Simple Function (SFNone) in key performance indicators important for IoT applications. MSF's improved network responsiveness and reliability, coupled with energy efficiency, make it a promising candidate for future 6TiSCH deployments in the IoT sphere. These findings suggest a potential shift towards MSF for enhanced IoT communication. For future research, investigating MSF's scalability and adaptability in larger and more dynamic network conditions would be beneficial. This could further establish 6TiSCH as a robust standard for the ever-growing IoT domain.